# Christopher Herbig

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## Education

**Ph.D. in Mathematics** **Graduation: May 2026**

Northern Illinois University, Dekalb, IL
Advisor: Dr. Michael Geline
Dissertation: *On Values Taken By Characters of Finite Groups*

**M.Sc. in Mathematics** **August 2021 – May 2024**

Northern Illinois University, Dekalb, IL
Specialization: Pure Mathematics
GPA: 3.959

**B.Sc. in Mathematics** **August 2018 – May 2021**

Northern Illinois University, Dekalb, IL
Program: General Mathematics
GPA: 3.941
Honors: Magna Cum Laude
Minor: Computer Science

## Research

### Interests

- Characters of finite groups and their values
- Minimal vanishing sums of roots of unity
- Permutation groups

### Preprints & Publications

- Herbig, C. On a question of Navarro on the field of values of characters of solvable groups, arXiv preprint (2026). (Currently under consideration for the *Journal of Algebra*)
- Herbig, C. A determination of $B$-groups of order $p^4$, arXiv preprint (2026). (Currently under consideration for the *Journal of Group Theory*)
- Herbig, C. and Hung, N. N. Character Values and Conductors of Low-rank Groups of Lie Type, arXiv preprint (2026).
- Herbig, C. Generalized Characters and Prime Graphs, *J. Group Theory*, to appear (2026).
- Herbig, C. Answer to a question of Hung and Tiep on conductors of cyclotomic integers. *Arch. Math.* **126**(3) (2026).
- Herbig, C. A Block-theoretic Proof of Burnside's Normal $p$-complement Theorem, arXiv preprint (2025).

### Conference Talks

- Minimal Vanishing Sums and Characters of Groups. Delivered a contributed talk in honor of Robert Boltje's 65th birthday at University of California Santa Cruz, Santa Cruz, CA, August 5, 2026.
- Conductors of Cyclotomic Integers and Fields of Character Values. Delivered an invited talk at AMS Sectional Meeting, Hartford, CT, April, 2025.

**Conferences Attended**

- 2026 Zassenhaus Groups and Friends Conference. Attended conference at Kent State University, Kent, OH, May 2026.
- Group Theory and Number Theory: Interactions. Attended conference in honor of P. H. Tiep's 60th birthday at Princeton University, Princeton, NJ, October 2023.

## Teaching Experience

**Assistant Professor of Mathematics** **August 2026 – Present**

Little Priest Tribal College, Winnebago, NE

- Full responsibility instructor for technical math, math readiness, college algebra, and precalculus.
- Advisor to liberal arts students, guiding course selection, career pathways, and internship opportunities.
- Designer of syllabi for all math courses to ensure compliance with the Nebraska Transfer Initiative.

**Graduate Teaching Assistant** **August 2021 – May 2026**

Northern Illinois University, Dekalb, IL

- Delivering recitations for Business Calculus (MATH 211), Introductory Discrete Mathematics (MATH 206), and Topics in Mathematics (MATH 101).
- Facilitating in-class discussions through the use of inquiry-based learning in College Algebra (MATH 110).
- Assisting instructors and professors to design, grade, and proctor tests for various courses.
- Tutoring undergraduate math students of all levels in NIU's Math Assistance Center.

**Adjunct Faculty** **August 2024 – May 2026**

Kishwaukee College, Malta, IL

- Full responsibility instructor for three sections of Topics in Mathematics (MAT 101). Each of these three sections was paired with a corequisite course (MAT 041) to further support students testing below the threshold for MAT 101.
- Full responsibility instructor for two sections of College Algebra (MAT 150) including one corequisite section (MAT 045).

## Internships

**Quantum Computing Research Intern** **May 2025 – July 2025**

Fermilab, Batavia, IL

- Collaborated with Dr. Henry Lamm's team to generalize an algorithm for synthesizing quantum gates.
- Developed C++ code which implements the generalized algorithm.

## Skills

### Programming Languages

- Proficient in C++ and FORTRAN
- Experience with Python and Julia.

### Software

- Proficient in typesetting documents with LaTeX.
- Experience with mathematical software such as GAP (Groups, Algorithms, & Programming).
- Experience with various online homework management systems such as Pearson's MyLab Math, Hawkes Learning, and ALEKS.
- Experience with learning management systems such as Blackboard, Canvas, and D2L Brightspace.

### Professional Development

- Teaching Assistant Institute participant at NIU (2021–2025)
- NIU Graduate Colloquium Committee member (2025–2026)
- Professional training in online course design for higher education through Quality Matters (2026)
- Attended Profesional Development & NTI Day at Central Community College, Grand Island, Nebraska (18 Sep. 2026).

## Honors

### Scholarships

- Donald R. and Rosemarie Armstrong Ostberg Memorial Fund in Mathematics (2025)
- Peter Nicholls Graduate Scholarship in Mathematical Sciences (2023)
- Norma K. Stelford Memorial Mathematics Endowment (2020)
- Tatsuji and Mami Kambayashi Memorial Scholarship (2020)

### Awards

- Outstanding Graduate Student Award (2025)
- First prize in 34th Annual NIU Math Contest (2021)

## References

References are available upon request.